UNIX Command

Basic Frequently Used Commands

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Example of a Unix/Linux command line interface (shell)

jjm@John-Desk2: ~/demo

```
#! /bin/bash
#
echo
echo "You invoked the script using the name  $0"
echo "There were  $# arguments. They are: " $@
echo "The first two args were: \"$1\" and \"$2\""
echo
echo "The complete list of args one per line (using \"\$@\") is:"
for arg in "$@"
do
  echo "\"${arg}\""
done
echo
echo "Using \$@ without the quotes doesn't do the same thing:"
for arg in $@
do
  echo "\"${arg}\""
done
echo
echo "It acts as if you'd used \$*"
for arg in $*
do
  echo "\"${arg}\""
done
echo
echo "But not as if you'd used \"\$*\"!"
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```
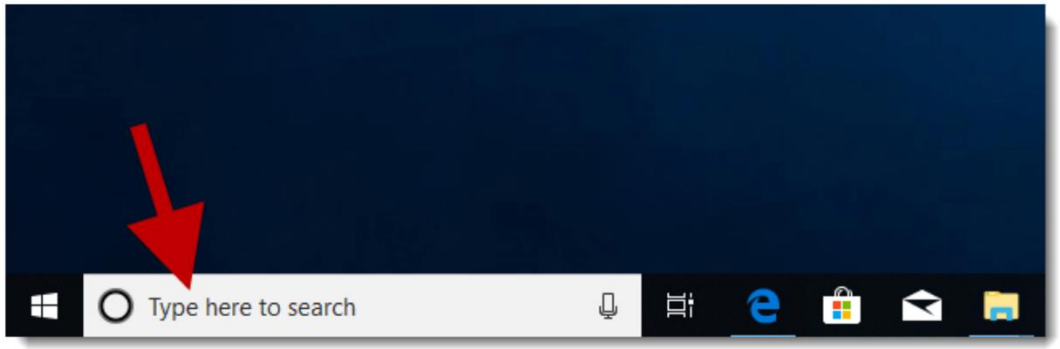

Accessing your local terminal or command prompt

- **Mac OS**
- Press ***command + spacebar*** -> ***Spotlight Search*** -> type ***Terminal***

- **Windows 10 or greater**
- type ***cmd*** in the search box (lower left), then choose the highlighted ***Command Prompt***



What is Unix?

- Unix is another proprietary ***operating system (OS)***
  - like Windows or MacOS (Unix based)
- There are many Unix and Unix-like operating systems, including open source Linux and its multiple distributions
- To work on the ***command line***, we need to be able to issue ***Unix commands*** to tell the computer what we want it to do

Ubuntu, Debian, AWS

cd -> change directory

How much Unix do we need to learn?

- **Navigating the File System:** Understanding the ***hierarchical structure*** of ***file directories***, using the `cd` command to move between directories.
- **File Paths:** Learning how to specify the location of files using ***absolute*** and ***relative paths***.
- **Basic Unix Commands:** Getting used with common commands like:
  - `ls` for listing files,
  - `mv` for moving files,
  - `rm` for removing files,
  - `mkdir` for creating directories,
  - `cat` for viewing file contents
- **Command Customization:** Learning how to modify the behaviour of Unix commands using `ls -l` to list files with detailed information

absolute path -> full path

File & Directory Management

Task	Command
Print working directory *Where are you now*	`pwd`
Change directory	`cd folder`
List files	`ls`
Long list (permissions) *rwxrwxrwx*	`ls -l`
Show hidden files	`ls -a`
Up one level (***back to the previous directory***)	`cd ..`
Home directory *1. user 2. group 3. others*	`cd ~` *relative path - reduce typing process*
Make directory	`mkdir folder`
Remove empty directory	`rmdir folder`
Remove file	`rm file`
Remove folder (***recursive -> forcing it***)	`rm -r folder`
Copy file	`cp file1 file2`
Copy folder	`cp -r folder1 folder2`
Move / rename	`mv old new`

Be careful: 'rm -r' will permanently delete your folder

View File Contents

Task		Command
View entire file		`cat file.txt`
Scroll through file	*type "q" to quit the view*	`less file.txt`
First 10 lines		`head file.txt`
Last 10 lines		`tail file.txt`

What if i want to view first 20 lines?
- head -n 20 filename.txt

Permissions

chmod -> change mode

Task	Command
Change permission	`chmod 755 file`
View permissions	`ls -l file`

d -> directory / folder
- -> file

File Permissions

chmod *octal file* – change the permissions of *file* to *octal*, which can be found separately for user, group, and world by adding:

- 4 – read (r)
- 2 – write (w)
- 1 – execute (x)

Examples:
chmod 777 – read, write, execute for all
chmod 755 – rwx for owner, rx for group and world

user group world

```
-rw-r--r--@ 1 bernardlee  staff   190358
-rw-r--r--@ 1 bernardlee  staff  2451463
-rw-r--r--@ 1 bernardlee  staff   926889
-rw-r--r--@ 1 bernardlee  staff  1004183
-rw-r--r--@ 1 bernardlee  staff     7789
-rw-r--r--  1 bernardlee  staff       31
-rw-r--r--@ 1 bernardlee  staff      165
-rw-r--r--@ 1 bernardlee  staff      165
```

6 4 4

Search Commands

Task	Command
Search text inside files	`grep "text" file`
Recursive search (*search* ***inside all files in this folder and its subfolders***)	`grep -r "text" .`
Find files by name	`find . -name "*.txt"`

Notes:

1. the **dot** (**.**) -> refers to ***this current folder***
2. ***asterisk*** (*******) -> ***wild card character*** to represent everything

if you want to kill the job

System Commands

Task	Command
Who am I?	`whoami`
System info	`uname -a`
Running processes *to quit -> type 'q'*	`top`
Disk usage	`df -h`
Folder size	`du -sh folder`
Command history	`history`

we want to copy all the command that we executed before?

- history > all_commands.txt

- history | grep "head -n"

"|" -> this is called pipe -> feed forward

Network Commands

To stop the ping process -> type 'ctr + c'

Task	Command
Ping a server	`ping google.com.my`
Download/request URL	`curl https://example.com`

curl --version

Example unix code:

Download and keep original filename (capital O)

1. **curl** -O https://www.gnu.org/licenses/gpl-3.0.txt

If curl is not installed
- sudo apt update
- sudo apt install curl

DOWNLOAD & RENAME FILE (small o)

2. **curl** -o gpl_page.html https://www.gnu.org/licenses/gpl-3.0.txt

DOWNLOAD WITH PROGRESS BAR

3. **curl** -# -O https://www.gnu.org/licenses/gpl-3.0.txt
4. wc -l gpl-3.0.txt

wc -> word count
-l -> how many lines

Zip & Tar

Task	Command
Create tar.gz	`tar -czf file.tar.gz folder/`
Extract tar.gz	`tar -xzf file.tar.gz`
Zip folder	`zip -r file.zip folder/`
Unzip	`unzip file.zip`

Side-by-Side Comparison

Feature	`tar -czf`	`zip`
Creates archive	✅ Yes	✅ Yes
Compression	gzip	zip
Default on Linux	✅ Yes	❌ No
Windows compatible	❌ Needs tool	✅ Native
Preserves permissions	✅ Yes	⚠️ Limited
Common file	`.tar.gz`	`.zip`

Other Useful Commands

Task	Command
Print text	`echo "hello"`
Clear terminal	`clear`

Example unix code:

echo -> display

1. name=Bernard
2. echo $name

Write text into a file

3. echo "Unix is fun" > file.txt

“>” means output the file

Append text to a file

4. echo "Hello world" >> file.txt
5. cat file.txt

“>>” means append / sambung

For more Unix Command

- https://tinyurl.com/5n6mwwsk

Thank you!!!